School of Electronics and Computer Science ELEC6200 MEng Group Design Project

GDP Team Number: 47
Title: See it, bite it, grip it, don't fall –using neurology to build a better elder/athlete
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Introduction

There are 3 main systems which contribute to a person's balance: vestibular, visual, and proprioception. Proprioception is the ability for a person to know where their own body parts are, and this is aided by the person's vision, and this along with their vestibular system helps them keep a reference to the rest of the world, giving the brain enough information to keep balance. If any of these 3 systems worsen through disability, disease, damage, or even just aging, this can severely affect a person's quality of life. For athletes, there is also an interest in improving balance above normal levels as this can also impact their strength and overall performance.

Group Design Project 47 was to use neurology to develop a prototype smart chair which could help improve both balance and strength of the user. The project consists of a chair with force sensors mounted onto the armrests, headgear for tracking head movement and jaw clench, wearable straps capable of vibration, and an LED display to stimulate eye-movement. When accompanied by an expert, a user would be able to use this chair to perform standard tests to improve balance.

For this group project, I stepped up as project manager, and in addition took up tasks from hardware, and supported software and firmware tasks where possible. As project manager, my main responsibilities were to plan tasks, assign time and people to them, and to adjust support for tasks as the scope and difficulty changed. In addition to this, I organised meetings with our team, the supervisor, and the client.

Since my background before this project was Mechatronics, my skills for supporting the project were mostly hardware focused, so these became the tasks I would take up. Since Mikkel had already worked on parts of this project prior, he was given highest priority in hardware tasks pertaining to his work, and I would take up the tasks which involved new components being added such as the EMG sensor and Vibration stimuli. In addition, I am also very competent with Arduino and MATLAB – two programming platforms which would be used for firmware and software. This meant that I was able to help kick-start the software and firmware sides of the project with my own help to get team members up to speed on using them, along with more technical assistance towards the end.

Project Management

In the first week of the project, it is important to identify strengths and weaknesses in the team, and the requirements for the project. I appointed 'section leaders'. I decided we would split the project into 3 different major sections which would need to be completed – hardware, software, and firmware. I then appointed a member of the team to take responsibility for each section's completion. This would primarily bring out people's individual skills, and secondarily helps push the project forwards faster as tasks can be subdivided easier. Since the team has 5 members, and only 3 sections to lead, Dan and myself took supportive roles in the team, filling in where other team members required and where tasks were too big to be taken by just a single person.

One of the early tasks completed as a team was to review what the project was, do background research, and create a specification of the project, and since we would want lab access, we also completed a risk assessment of the project to the specification. The ordering of hardware was started very early too – I estimated that due to the coronavirus pandemic, it is likely that hardware components could take longer than usual to arrive – we estimated delays up to a month. The project plan included this delay so that tasks which would not require hardware would be completed first, and tasks requiring the hardware would come later.

From the first week onwards, I organised every meeting, with the goal of setting a regular schedule of meeting on Tuesdays around 2pm, with extra weekend meetings if required. Agendas for these meetings typically were limited to around 4 topics with a final topic of "Any Other Business". This duration of meeting tended to work for roughly 30 minutes to 1 hour long meetings and helped push the dialogue of the meeting past anything unnecessary, so that everyone could focus fully.

In addition to meetings, tasks were organised on a weekly basis, with each meeting finishing off with a confirmation that everyone in the team was happy and comfortable with the task they had been assigned that week, and the deadline. Each of these tasks was also assigned within an 'epic' – which was a useful way to organise tasks together so that it was clear what stage of the project we were in.

Kanban Project Management

For this project, since it had already been started, there was already a goal in mind and some of the work had already been started. These two factors lead me to decide to do a Kanban management system for a more structured approach to management rather than a more iterative system such as Agile management.

Kanban management sorts all work into 3 categories: To Do, In Progress, and Finished. Each of these sections can be further split into sub-sections. The entire board of work then is split into different 'epics', such as Setup, Hardware, Software, etc. which lets us track the progress of each part of the project. Combining this with JIRA to track all of the tasks allowed very easy analysis of the state of the project, remaining tasks, people assigned to each task, and what rate the project was progressing at.

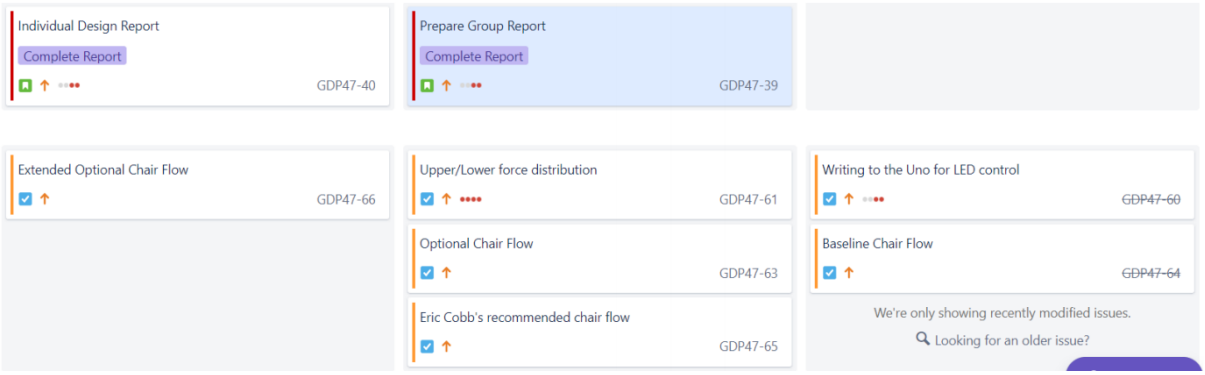

Figure 1 Kanban Board to organise tasks

Another important task as project manager was to analyse risk of the tasks. This would mean going through every task on the Gantt chart and evaluating both how likely from 1-5 a consequence could happen, and from 1-5 how big of an impact that would have on the project. Once a score had been assigned to the risk of that task by multiplying these two factors, appropriate control measures could be planned. By starting this early in the project, it allowed us to foresee potential difficulties and to work around them so that the project could be completed whether or not difficulties arise from tasks falling behind or becoming impossible. The main method of mitigation chosen for most low-risk tasks was to have ample extra-time at the end of the project, ready to be filled up if tasks delay the project or if new tasks need completing, but also by spreading out the critical path so that high risk tasks do not take up the critical path, it ensures that high risk tasks are capable of crashing by allowing dynamic re-allocation of project resources (time and people assigned) to catch the tasks up.

Task re-planning

As I estimated, many of the hardware tasks were delayed due to supply issues caused by the pandemic. This was however mitigated as planned by re-assigning the tasks so that the project plan was more nonlinear, which allowed tasks to be completed as components arrived and where possible – and by supporting the software and firmware teams to begin simulation of data so their tasks can speed along as fast as possible.

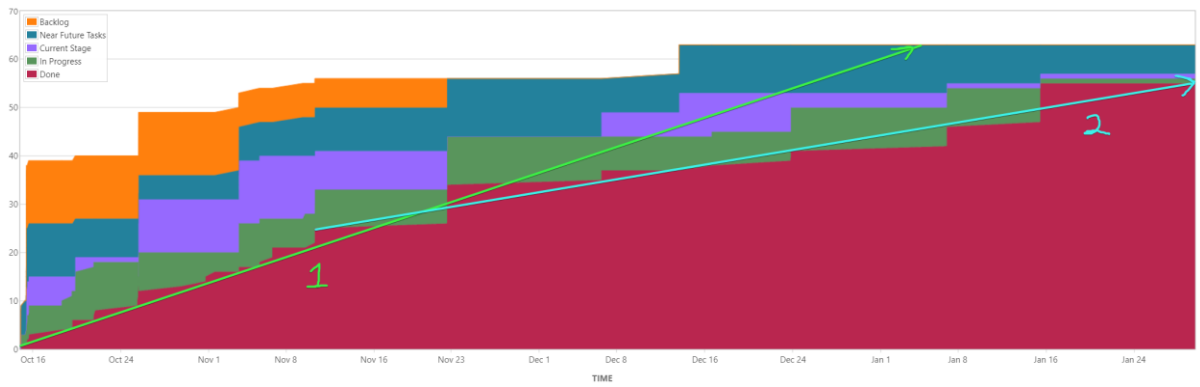

Figure 2 Cumulative Flow Chart with predictions labelled – 1 being before the prediction before the delays and rescheduling and 2 being new work velocity after delays and rescheduling slows the progress

Technical Contributions

As the project manager and having experience in all of the technical aspects of the project, most of the tasks I took up were tasks to help support others in the project where needed. There were 2 major tasks which I took most of the work for: the EMG sensor, and the vibration stimuli

EMG Sensor

Before starting hardware tasks, it is essential to plan what could be required and research what components are appropriate. The task of the EMG sensor was to detect jaw clench – we considered using a force sensor such as a load cell to detect strength of biting down, or an EMG sensor to detect the nerve activity. Since it is significantly less intrusive and more hygienic, an EMG sensor was chosen. This decision was made very early on in the project in the week starting 12/10.

For an EMG Sensor the best option for a project like this was the Myoware Muscle Sensor by Sparkfun. This sensor is capable of measuring nerve activity for driving muscles, and this sensor is built for outputting this to an Arduino, making it a perfect choice.

The EMG sensor arrived, and testing began in the week starting 9/11. Testing was easy to set up by creating a simple program in the Arduino IDE to read the data currently detected by the sensor and display it on the screen. This functionality was verified by attaching it to a large muscle – my bicep – and seeing what the range of values was when tensed vs untensed. The data read was then output into MATLAB and saved as a file to be sent to the software team to ensure that they were able to model it appropriately.

The final question was whether or not the system would require any filtering or amplification, however upon checking with the software teams it was decided that it would not be necessary. This allowed the final stage to be completed where a permanent connection was soldered onto the EMG sensor, and a circuit created for the headpiece where the EMG sensor could easily be attached to.

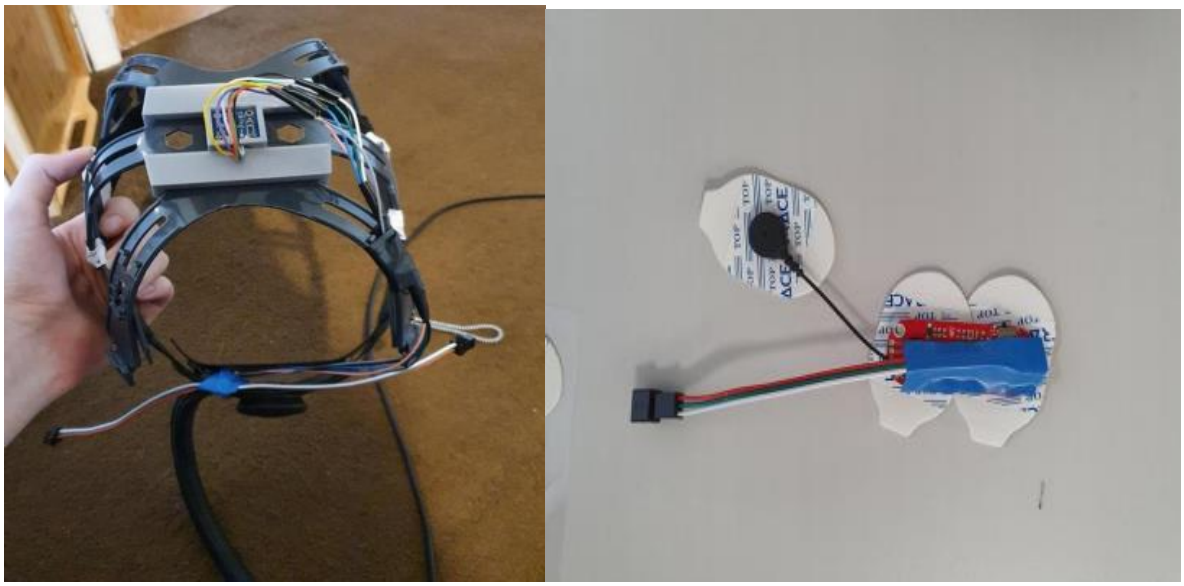

Figure 3 EMG sensor and headpiece

Vibration Motors

Once again, the success of this task would rely on proper planning. The most important feature of the vibration motors was the range of frequency. An initial value of 2-1000Hz was given, however from deeper research 50-400 Hz was a more specific range for the uses we needed. In the first client meeting in the week of the 19/10, our team clarified with the client what range would be appropriate.

Now that a good range had been found, several solutions were discussed between me and Mikkel after the EMG sensor tasks were completed (the week 16/11). The proposed ideas were a piezoelectric buzzer, a speaker, or an actuator with a rubber tip. It was possible to test 2 of these ideas in labs, however the buzzer and speaker were obviously not appropriate solutions when tested as they were far too loud and did not cause enough stimulation.

When doing a second round of research to find appropriate solutions for actuators, the solution of ERM motors was found. This would limit the range down to 50-300 Hz, which was approved by the client 2 weeks later after we explained the limitations in our client presentation. While waiting on verification from the client, the motors were so cheap that I decided to place an order for them early in case shipping was delayed, and so that we could be sure they would work for the client meeting. Upon testing them it was obvious they would be a success.

Unfortunately, to be able to reach a range of 50-300Hz, 2 motors would be needed since one motor did not give a full enough range. This turned out to be an easy solution however, by just placing both motors next to each other. The more difficult part was making the motors wearable.

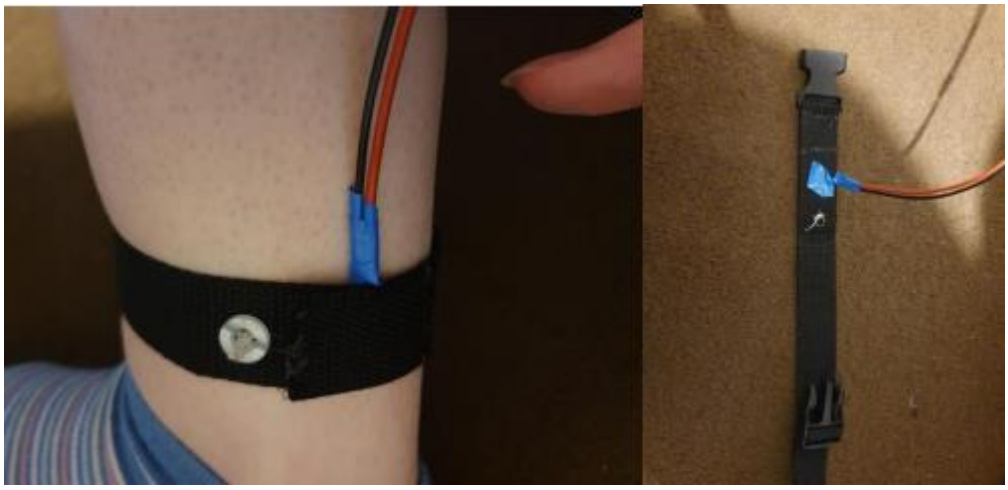

Figure 4 Vibration motors attached to a wearable strap

Since a core design consideration of this project was to be accessible to the elderly, and to any body shape, it was important that these motors would be adjustable to the user. Secondly, they needed to not be intrusive. During the second meeting the client reassured us that the placement was good– the back of the neck, jaw, and ankles. Later however, we realised that the jaw would be inaccessible without glasses to attach them to, and since the eye-tracking glasses portion of the project fell through this idea got left out. While ankle straps for vibration were completed, the necklace created couldn't be finished as LED display tasks took priority for the remainder of lab time.

LED Display

As Mikkel had returned home for Christmas and would be unable to continue working physically on the project due to coronavirus restrictions, the LED display in its unfinished state would be left in

labs. Unfortunately, this part of the project had been worked on with little communication and had been included as a part of the project mid-way through, so included a mains connection which was unsafe to use. This meant that before the project could be taken out of labs for Christmas, a full new risk assessment, risk mitigation strategy, and implementation of that strategy had to be completed in the remaining weeks (30/11 + 7/11). This involved working with Jeff Hooker in labs to identify risks of the LED display and insulate it well enough to pass a PAT test and provide enough insulated inputs and outputs for it to still power the LED display and have communication via an Arduino and Bluetooth module. Once this component had been made safe in labs, it had to be transported and then repaired, and improved so that the Bluetooth module was properly connected (14/12) so that testing could begin.

Testing

Since I was the team member who took as much hardware as I could home to continue the project, all testing was left to me to complete with support from the rest of the team. This meant checking and improving every hardware connection, trialling all code from the lowest firmware level implemented up to generating data to test high level software, and then completing full tests of the chair-flow to check that it is appropriate and accessible to use.

Testing often involved a long process of finding errors and working with the programmers to implement code to error-handle bugs, and to output debugging statements to be able to locate where errors were happening, and finally logic-checking others code as a second opinion to see whether or not any overall features (such as scrolling the menu left or right) required different methods.

Conclusion

I feel that I was able to add a lot of value with my experience in the technical aspects of the project to make big improvements. I also believe that as the project manager I made appropriate assessments and managed the unusual circumstances appropriately and managed to motivate the team to produce a good quality of result despite the challenges. The scale of the project made it difficult to manage the time everyone could spend working in labs due to the lab restrictions, and the time needed for meetings and non-practical work. Despite this however, I feel particularly proud of my accomplishments as project manager: I never required working on weekends if team members were uncomfortable with it, I never required unreasonable hours of crunch even towards the end of the project, and despite reduced lab access I still ensured that everyone had the opportunity to see the project in person before labs closed. Most importantly, I feel accomplished that the core features as requested by the client were met by the whole team, with other parts of the project being completed to an even higher standard than requested.

Appendix

Name:	Alexandra
Date:	5/10/2020
Tasks: N/A – early week so tasks not assigned yet	
Decisions: Step up to be group leader. Should organise a get together quickly so that work can begin as soon as possible. Make sure that everyone is able to introduce themselves so that the group can start to be friendly and tasks can start to be figured out.	
Work done: Written emails to everyone in the group and set up the Team server for the team to communicate in. Organised an in-person meeting so that we can all meet face-to-face, after checking with the team multiple places they would feel comfortable to meet at, and where regulations allow us to meet. Organised agenda and tasks required for people to complete at the meeting.	
Comments:	

Name:	Alexandra
Date:	12/10/2020

Tasks:
Create Gantt Chart, organise meetings, Background Research.

Decisions:
Judgement calls on every task's duration and difficulty, along with assessing the risk of every task.
Choose to delegate to different people based on what their preferences were and their skills.
Decide how complicated the Gantt chart should be so that it is sufficiently informational, along with being usable enough that every team member is likely to regularly check it.

Work done:
Set agendas for meetings, signed up to ZHealth's neurofundementals course and looked through lectures, read background research from Zotero which was set up by Mikkel, created Gantt chart.

Comments:
Tasked the team to create a list of tasks that they would want to do and what they think is necessary, and how long they think it will take.
Components started to get ordered here.

Name:	Alexandra
Date:	19/10/2020

Tasks:
Organise client meeting, prepare presentations.
Set up JIRA page and roadmap.

Decisions:
Based on work process so far, and experience in the past, JIRA was chosen as a good place to issue and track tasks from as it was clear that excel management was not fully appropriate for issuing deadlines and tracking tasks due to the slow rate people checked the spreadsheets at. Jira integration helps manage everything properly.
Decide which people will be doing which sections of presentations, and internal deadline for presentations

Work done:
Created a list of every task to be completed, assigned each task and created sub-tasks to individualise the work assigned even in tasks which required more than 1 person. Held instructional meeting to teach team how to use JIRA site, and created JIRA roadmap.
Sectioned up presentation and held meeting to ensure that everyone was able to understand how to work on the presentation, worked on my own slides as an example, and then held another meeting to rehearse presentation, and also arranged meeting with a client to also show the presentation to.

Comments:
Internal presentation deadline was missed by a couple days, but still finished on time for the actual deadline.

Name:	Alexandra
Date:	26/10/2020

Tasks:
Begin Hardware, Firmware, Software intro tasks.

Decisions:
Components have not arrived yet, so most hardware assigned to me cannot be started yet. Instead, decide to delay my hardware tasks and begin by assisting Zee with setting up MATLAB environment.

Work done:
Held meetings with Zee to set up MATLAB environment and teaching how to create GUIs in MATLAB to get him up to speed with programming in MATLAB. Continued background research into hardware components required, filtering and amplification.

Comments:

Name:	Alexandra
Date:	2/11/2020

Tasks:

Re-plan project tasks to mitigate risk from hardware being late.
Coordinate with team to help recover tasks which are falling behind.

Decisions:

Chose to get support from team to make decision to re-plan project, taking stock of all tasks required yet to do, and all progress done so far to decide which tasks can be done while we wait for hardware, and how to catch up from immediate delay. Organised to swap to simulation tasks so that software and firmware tasks can begin progress despite hardware delays.

Work done:

Worked with hardware components currently in labs to do hardware tasks which were currently possible and held more meetings with zee to talk about GUI and making decisions about the process of chair-flow of the project.

Comments:

Name:	Alexandra
Date:	9/11/2020

Tasks:
Begin work on accelerometer circuit and EMG sensor in lab. Create final hardware list

Decisions:
Made decision over whether or not default filtering on the EMG sensor is appropriate for our use and the range of frequencies appropriate for accelerometer.

Work done:
Planned for lab time by researching accelerometer and EMG sensor data sheet. While in lab, put together EMG sensor and wired it up to Arduino. Wrote code to test it out and print output. Sent output to team to check that they are ok with the range of data received and sensitivity, verified functionality. Held meeting with Mikkel to decide accelerometer functionality.

Comments:

Name:	Alexandra
Date:	16/11/2020

Tasks:
Complete EMG sensor & Test. Trial vibration stimulus.

Decisions:
Decide whether or not vibration methods were fit to test – trialled 2 different methods and deemed both nonviable.

Work done:
Finalised EMG sensor circuit and began work in labs on vibration tests. Researched other methods and found ERMs for causing vibration. Held meetings with team and supervisor and prepared explanation and questions for client to discuss whether or not the resultant vibration method is appropriate.

Comments:

Name:	Alexandra
Date:	23/11/2020

Tasks:
Test ERMs, create straps, begin final preparation of testing. Test full workflow of hardware to firmware to software.
Run meetings to finalise chair flow.

Decisions:
Found a way to make motors wearable, instead of deciding specifications of size, decided to make it adjustable since no compromise was appropriate.

Work done:
Once new motors had arrived, began testing in lab. Once functionality was verified, got straps so that they could be sewn into wearables. Worked on creating prototype with wearables for ankles since they were the easiest to create and test. Installed 1 motor and connected it for each leg. Took it into lab and tested.

Comments:

Name:	Alexandra
Date:	30/11/2020

Tasks:

Create client update email, work on demonstration video and catalogue currently existing work and remaining tasks for December push.
Accelerometer head-sway meetings.
Risk mitigation strategy for LED display

Decisions:

Software is falling behind – receiving feedback that to progress faster, data and testing is needed. Decide to make sure that as many team members as possible are able to make it into lab before the labs likely close. Decide to make a sub-team for software since software task order is best managed and will best meet deadlines when managed by those who understand the code better in its current state. Team will report back to me to ensure that tasks are not falling behind and can dissolve after Christmas break.

Work done:

Begin connecting as much hardware together as possible so that readings can be collected to send to software team for using as example data. Create a demonstration video once everything is in labs, catalogue with team all work that is required to be completed over December so that redundant tasks can be removed to ensure project finishes.
Assign sub-team for software now that data has been collected and hold meetings to ensure that they are equipped to solve their tasks.

Comments:

Dan & Aaron & Zee team forms
LED display put into labs but is using mains which was not originally in the plan or risk assessment, so the risk assessment has to first be updated to ensure that risk is mitigated, and then it can be fixed to be safe to use.

Name:	Alexandra
Date:	7/12/2020

Tasks:

Fix & make safe LED display. Move as much equipment as possible back home for Christmas break so that testing, and completion can continue.
Repair any damages that might happen to the project throughout transport.
Run meetings & begin work on group report.

Decisions:

Decide to delay vibration motors as LED display is an extremely important part of the project and must be fixed immediately so as much can be taken out of labs before labs close.

Work done:

Organise meetings to get team on the same page about risk of LED display. Once it has all been signed off, go into labs and begin working on executing the plan to make the LED display safe. Once it has been made safe, pack up everything possible in the lab so that it can be taken home over break. In addition to this, run meeting with Mikkel to catalogue all hardware which is existing on the smart-chair to form a list of repairs which needed for the chair once home.

Comments:

Name:	Alexandra
Date:	14/12/2020

Tasks:
Support software and firmware teams in UI development, while continuing to improve hardware reliability.

Decisions:
Check in with software team regularly to ensure that they are keeping pace, decide to just set overall deadlines rather than micromanaging at this point in the project.

Work done:
Held a meeting with software team to begin working on UI redesign tasks and introducing new features such as the settings menu and demonstrating how to program popup menus and save data from them.
Continue work on repairing chair and mostly LED display and focus on fixing errors in wiring for the LED display.

Comments:

Name:	Alexandra
Date:	28/12/2020

Tasks:
Help with UI re-design now that UI is finalising and begin testing and debugging project.

Decisions:
Software tasks still lagging a bit behind so offer more help to speed those tasks up. Mostly make sure to get people focused on regular working meetings around midday on workdays so that people aren't wasting time.

Work done:
Sat in software meetings to help with UI redesign tasks, and focused on working meetings with Dan, Aaron and Zee to implement any changes anyone in the team suggests and test how they work. Offer debugging support for the code submitted as well so that the code runs smoothly. Start work on calibrating the force-plate so that debugging with a full trial can begin.

Comments:

Name:	Alexandra
Date:	4/01/2021

Tasks:
Continue testing & go through people's sections on group report with them.

Decisions:
This was the internal deadline for drafts of sections of the report, so many decisions around how to restructure report or move sections, assign bigger tasks to some people who have had sections which end up being pretty small and ensuring that those writing more aren't having to write too much.

Work done:
Re-calibrate chair to try to overcome issues with carpet, zero errors, and centre of pressure not properly lining up.
Continue debugging and testing of chair – with the regular working meetings so anyone in the team is able to contribute suggestions or changes if they notice anything which could help.
On my own, resolder many connections in the project to make it all a lot more reliable, and allow the project to be moved to have better cable-management for presentation.

Comments:

Name:	Alexandra
Date:	11/01/2020

Tasks:
Finalise project & Finalise group report

Decisions:
Prioritise final tasks which are required to make the chair work, and remove all excess tasks which are not essential that were being worked on to save time so that everyone is able to hand in their reports and complete the presentation in time. This causes the chair to ultimately not be perfectly calibrated and certain parts of the project are less polished than others – such as settings not persisting between closing and re-opening the program, but these are not more important than finishing the project on time.

Work done:
Finalise debugging software and collect final data for analysis and for the reports and presentations. Read through team's sections and format report with Mikkel to ensure that everything is there. Begin reading through people's individual reports and checking through everyone's work logs to make sure that everyone has been able to keep up with recorded dates – help people fill in missing sections.

Comments: